



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

B.Sc. PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	SCIENTIFIC DOCUMENTATION				
Type of Course	Vocational Minor (GROUP I: TECHNIQUES IN MATERIALS PHYSICS)				
Semester	VIII				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Basic computer operating knowledge				
Course Summary	Master the art of characterizing nanomaterials with microscopy, diffraction, and spectroscopy techniques, unlocking their secrets and exploring their diverse applications.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Create professional-quality scientific documents, including research papers, reports, and theses, using LaTeX typesetting system.	Ap	P	Practical Assignment / Observation of Practical Skills
CO2	Develop proficiency in formatting and structuring scientific content effectively, adhering to established conventions and guidelines	Ap	P	Instructor-created exams / Home Assignments
CO3	Gain skills in incorporating complex mathematical equations, figures, and	An	P	Practical Assignment /

	tables seamlessly into LaTeX documents to enhance clarity and understanding.			Observation of Practical Skills
CO4	Learn to manage citations and references efficiently using BibTeX or BibLaTeX, ensuring accuracy and consistency in academic writing.	An	P	Practical Assignment / Observation of Practical Skills
CO5	Acquire techniques for designing and delivering engaging presentations and posters for scientific conferences and academic events using LaTeX Beamer class.	Ap	P	Seminar Presentation / Writing Assignment
CO6	Develop collaborative writing and version control skills, enabling them to work effectively with co-authors and collaborators on LaTeX documents for scientific communication and publication.	E	P	Seminar Presentation / Group Tutorial Work
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I		INTRODUCTION TO SCIENTIFIC WRITING AND LaTeX	11	15
	1	Understanding the importance of clear communication in science Overview of the scientific writing process	1	
	2	History and purpose of LaTeX Setting up LaTeX environment (installing LaTeX distribution, editor)	2	
	3	Creating a simple LaTeX document Sections, subsections, and paragraphs	2	
	4	Formatting text (fonts, styles, sizes), special characters, text alignment	2	
	5	Writing mathematical equations and symbols	2	

	6	Arrays and matrices	2	
II	ADVANCED LaTeX TECHNIQUES		11	15
	7	Inserting figures and tables into LaTeX documents Captioning and referencing figures and tables	2	
	8	Explore different properties like rotate, scale, etc, wrap figures	2	
	9	Cross-referencing sections, equations, figures, and tables Managing citations with BibTeX or BibLaTeX	3	
	10	Customizing page layout (margins, headers, footers) Creating custom document classes and styles, numbering, footnotes	2	
	11	Generating bullet and numbered lists Customizing list styles	2	
III	CREATING PRESENTATIONS WITH BEAMER		13	20
	12	Introduction to Beamer class for presentations	2	
	13	Creating slides, adding frames	2	
	14	Dividing the slide into multiple columns, adding different blocks, etc	2	
	15	Table of contents	2	
	16	Overlays - Pause, Slide Transitions	2	
	17	Designing posters with LaTeX	3	
IV	WRITING SCIENTIFIC DOCUMENTS		13	20
	17	Structuring a research paper (abstract, introduction, methods, results, discussion)	3	
	18	Structuring reports, theses and books	2	
	19	Defining custom environments for specialized content Creating macros for frequently used commands	2	
	20	Understanding journal-specific formatting requirements Tips for submitting articles to scientific journals	2	
	21	Collaborating on LaTeX documents with co-authors and editors	2	
	22	Citing references and inserting the bibliography	2	
V	OPEN ENDED MODULE		12	
	Hands-on training to prepare some of the following documents, poster, presentation or any other relevant designs.			

1	Prepare a document presenting the mathematical proof of a theorem in physics		
2	Prepare a document showing examples of different matrix operations.		
3	Design a model question paper for this course		
4	Prepare a neat presentation using beamer demonstrating its various features.		
5	Designing a scientific posters for conferences and presentations		
6	Prepare a scientific paper for specific journal (Use the document class of Physical Review, Science Direct etc.)		

Books and References:

- 1) A Short Introduction to Latex: A Book for Beginners by Firuza Karmali Aibara
- 2) LaTeX: A Document Preparation System" by Leslie Lamport
- 3) The LaTeX Companion" by Frank Mittelbach, Michel Goossens, Johannes Braams, David Carlisle, Chris Rowley
- 4) LaTeX Beginner's Guide" by Stefan Kottwitz

Mapping of COs with PSOs and POs :

	PS O1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	0	3	0	0	0	0	2	2	2	0	0	0	0
CO 2	0	3	0	0	0	0	2	2	2	0	0	0	0
CO 3	0	3	0	0	0	0	2	2	2	0	0	0	0
CO 4	0	3	0	0	0	0	2	2	2	0	0	0	0
CO 5	0	3	0	0	0	0	2	2	2	0	0	0	0
CO 6	0	3	0	0	0	0	2	2	2	0	0	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1		✓	✓	✓
CO 2	✓	✓		✓
CO 3		✓	✓	✓
CO 4		✓	✓	✓
CO 5	✓	✓		✓
CO 6	✓	✓	✓	✓